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Communication

Catalytic Asymmetric Axially Chiral Allenyl C–P Bond Formation

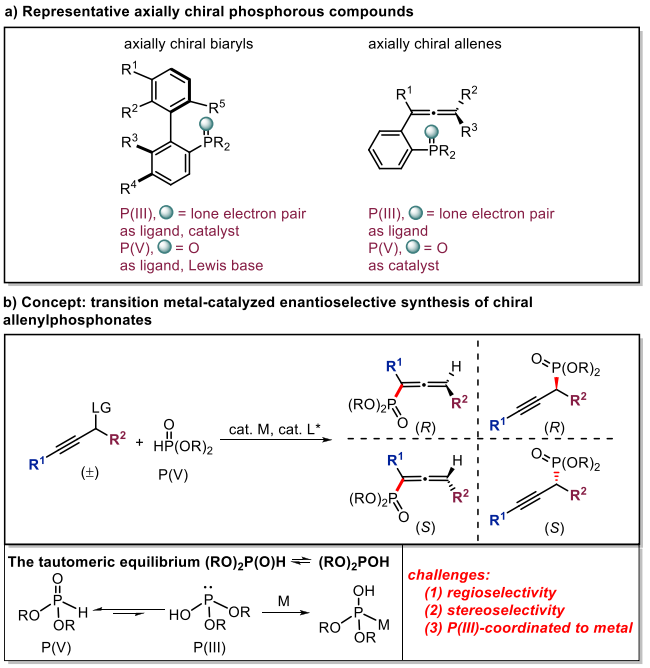
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ABSTRACT: Chiral organophosphorous compounds are very important in catalysis, organic syntheses, and medicinal chemistry. However, catalytic enantioselective protocols for the axially chiral allenyl phosphorus compounds have never been reported. Herein, a palladium-catalyzed enantioselective carbon–phosphorus bond formation reaction affording axially chiral allenyl phosphonates has been developed. The reaction enjoys high yields and ees accommodating a wide range of functional groups. Mechanistic studies have unveiled an overwhelming kinetic resolution process.

Chiral biaryls[1] and allenes[2,3] are two very common classes of most important axially chiral compounds. Biaryl bisphosphines have been extensively applied as ligands and catalysts for asymmetric syntheses.[4] In addition, although commonly used as Lewis bases,[5] some examples of biaryl bisphosphine oxides have also been reported as ligands in transition metal-catalyzed coupling reactions (Scheme 1a, left).[6] However, reports on axially chiral allenyl phosphorus compounds are very limited (Scheme 1a, right). Ready and co-workers had identified axially chiral allenylphosphines as ligands and catalysts for asymmetric syntheses.[7,8] So far, there has been no report on catalytic asymmetric synthesis of axially chiral allenyl phosphorus compounds.[9−11] The envisioned challenge is the tautomeric equilibrium between the pentavalent $(RO)_2P(O)H$ and trivalent $(RO)_2POH$, which may coordinate overwhelmingly with the catalytic amount of the palladium in the chiral Pd catalyst with a chiral ligand, inhibiting the reaction and/or excluding chiral induction (Scheme 1b, bottom left).[12] In addition, the regioselectivity of allenylic vs propargylic phosphonates is also an issue (Scheme 1b, top).[13]

Scheme 1. Representative Axially Chiral Compounds and Strategies for the Catalytic Asymmetric Synthesis of Allenylphosphine Derivatives



We initiated our study on the reaction of racemic 1-phenylhept-2-ynyl benzoate **1a** with diethyl phosphite **2a**. After tedious optimization, it was observed that the reaction worked best by using $Pd_2(dba)_3{\cdot}CHCl_3$ as the catalyst. A variety of commercially available ligands were examined with THF at 70 °C (Table 1). The reaction with axially chiral biaryls **L1**–**L7** as ligands afforded the allenylic products (*R*)-**3aa** in 24–92% yields; however, the enantioselectivity was only 24%. Switching the ligand to SKP (**L8**)[14] and SDP (**L9**)[15] with spiro-chirality, **L10** with planar chirality, **L11** and **L12** with central chirality, **L13** and **L14** with planar and central chirality, and Sadphos (**L15**–**L17**)[16] with central chirality, the ees were still low. We were pleased to find that the desired allenyl phosphonate (*R*)-**3aa** was obtained in 69% yield with 86% ee when (*R*,*R*)-DACH-Phenyl Trost ligand (**L18**)[17] was used as the ligand. Thus, a series of such Trost ligands with chiral diamine skeletons, **L19**–**L21**, were tested; however, no better result was observed. We were happy to see that excellent regioselectivity was achieved in these Pd-catalyzed coupling reactions without the undesired propargylic product being formed.

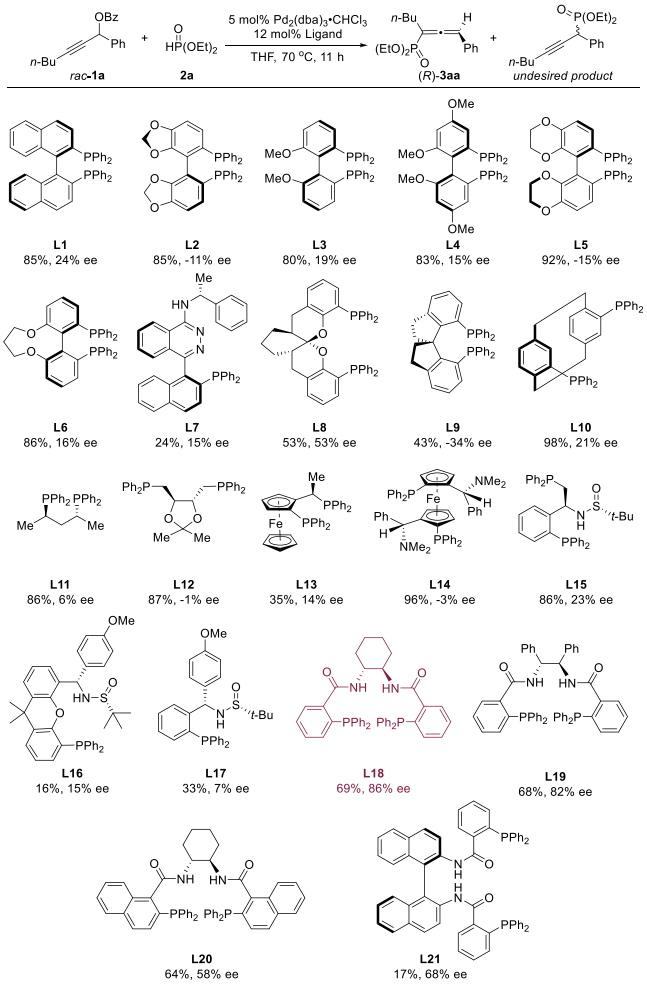
With the optimal ligand **L18**, the effect of the leaving group was evaluated, and the leaving group was found to have a profound effect on the enantioselectivity (Table 2). The

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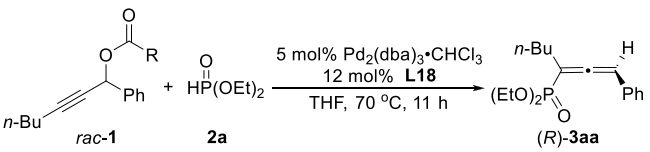
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Table 1. Ligand Screening[a]



[a]The reaction was conducted with **1a** (0.2 mmol), **2a** (0.1 mmol), $Pd_2(dba)_3$·$CHCl_3$ (5.0 mol%), and ligand (12 mol%) in THF (1 mL) at 70 °C under N_2 atmosphere. The yield was determined by ^1H NMR analysis, and the ee value was determined by HPLC analysis.

Table 2. Effect of Leaving Group[a]

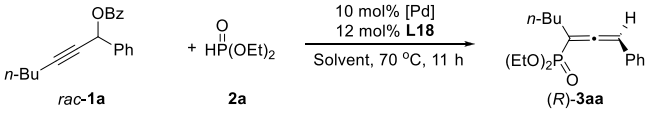


entry	R	recovery of **2a** (%)	NMR yield of (*R*)-**3aa** (%)	ee of (*R*)-**3aa** (%)
1	**Ph**	**21**	**66**	**90**
2	Ad	42	48	94
3	OMe	0	72	65
4	OBn	0	61	80
5	1-naphthyl	0	63	77
6	2-furanyl	16	27	74

[a]The reaction was conducted with **1** (0.25 mmol), **2a** (0.1 mmol), $Pd_2(dba)_3$·$CHCl_3$ (5.0 mol%), and **L18** (12 mol%) in THF (1 mL) at 70 °C under N_2 atmosphere. The yield was determined by ^1H NMR analysis, and the ee value was determined by HPLC analysis.

reaction of 1-phenylhept-2-ynyl adamantoate afforded (*R*)-**3aa** in 94% ee with a low yield of 48% (Table 2, entry 2). In addition, methyl carbonate (Table 2, entry 3), benzyl

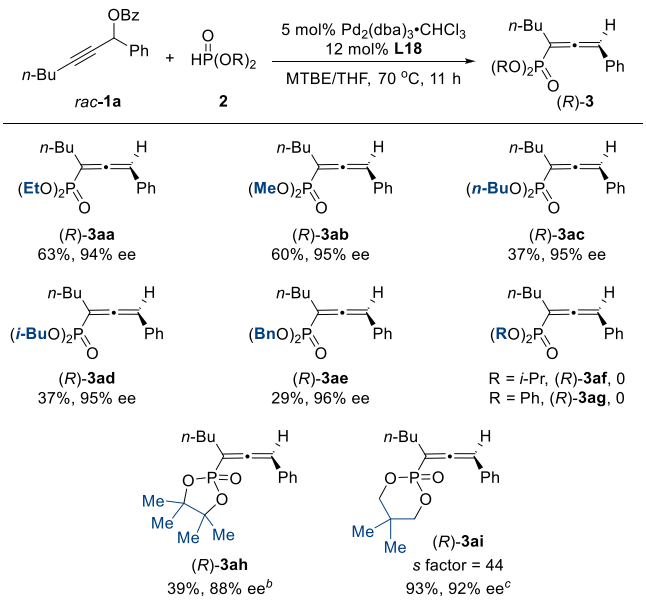
Table 3. Effect of Solvent and Palladium Catalyst[a]



entry	solvent	catalyst	recovery of **2a** (%)	NMR yield of (*R*)-**3aa** (%)	ee of (*R*)-**3aa** (%)
1	toluene	$Pd_2(dba)_3$·$CHCl_3$	15	31	91
2	MeCN	$Pd_2(dba)_3$·$CHCl_3$	15	27	4
3	dioxane	$Pd_2(dba)_3$·$CHCl_3$	29	55	86
4	MTBE	$Pd_2(dba)_3$·$CHCl_3$	28	55	94
5	DME	$Pd_2(dba)_3$·$CHCl_3$	27	46	90
6	CMPE	$Pd_2(dba)_3$·$CHCl_3$	21	41	90
7	c-Hex	$Pd_2(dba)_3$·$CHCl_3$	14	30	91
8	THF	$Pd_2(dba)_3$·$CHCl_3$	26	66	90
9	THF	$Pd_2(dba)_3$	28	54	91
10	THF	$Pd(dba)_2$	7	65	89
11	THF	$Pd(OAc)_2$	74	5	91
12	THF	$PdCl_2$	79	0	-
13	THF	$[Pd(allyl)Cl]_2$	49	14	84
14[b]	MTBE/THF	**$Pd_2(dba)_3$·$CHCl_3$**	**19**	**70**	**94**

[a]The reaction was conducted with **1a** (0.25 mmol), **2a** (0.1 mmol), [Pd] (10.0 mol%), and **L18** (12 mol%) in the solvent (1 mL) at 70 °C under N_2 atmosphere. The yield was determined by ^1H NMR analysis, and the ee value was determined by HPLC analysis. [b]MTBE/THF (0.8 mL/0.2 mL).

Scheme 2. Scope of *H*-Phosphonates[a]

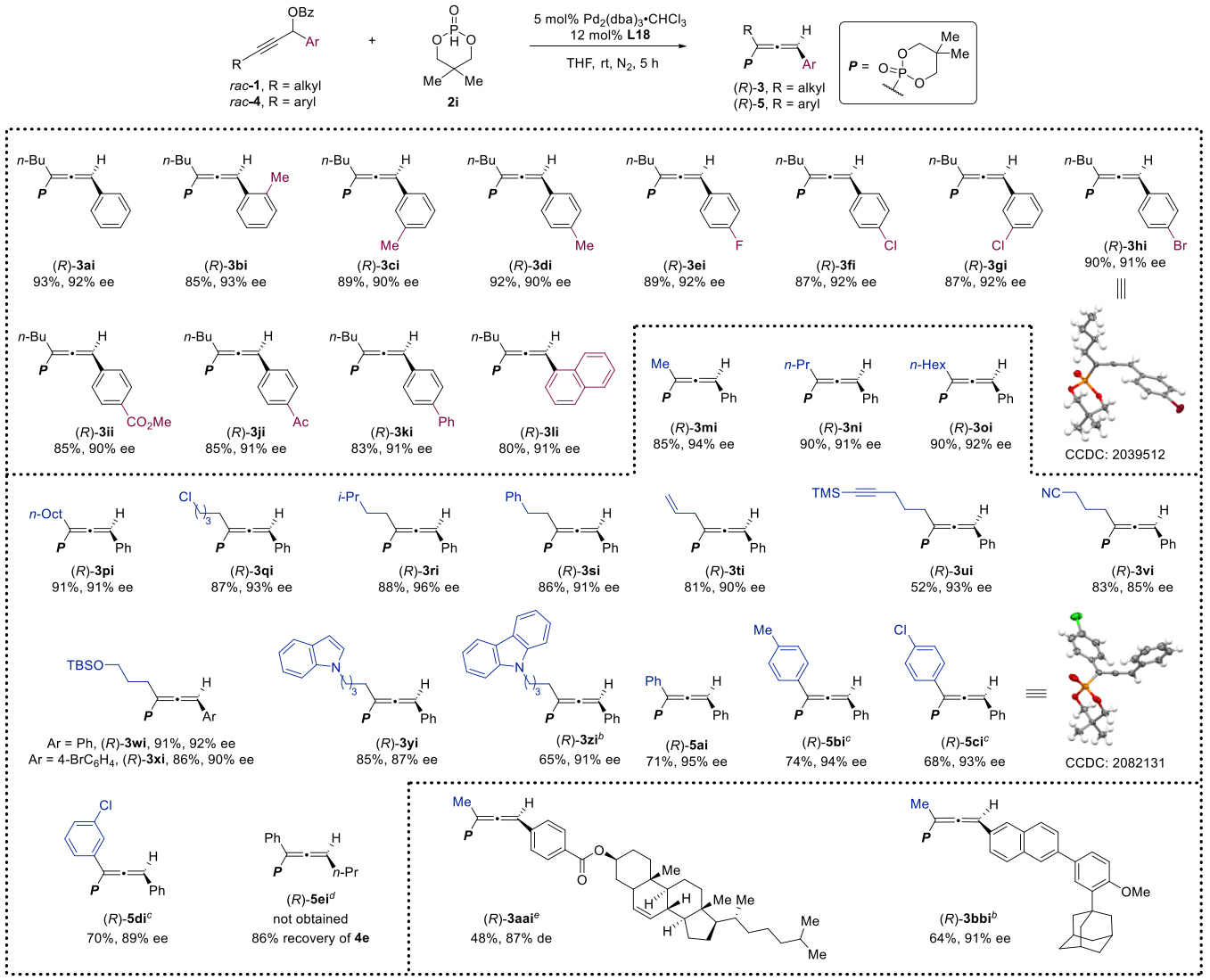


[a]The reaction was conducted with **1a** (1.25 mmol), **2** (0.5 mmol), $Pd_2(dba)_3$·$CHCl_3$ (5 mol%), and ligand (12 mol%) in MTBE/THF (4 mL/1 mL) at 70 °C under N_2 atmosphere. The ee value was determined by HPLC analysis. [b]MTBE as solvent, rt, 5 h. [c]THF as solvent, rt, 5 h.

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Scheme 3. Scope of Propargylic Benzoates[a]



[a]The reaction was conducted with **1** (1.25 mmol), **2a** (0.5 mmol), $Pd_2(dba)_3{\cdot}CHCl_3$ (5.0 mol%), and **L18** (12 mol%) in THF (5 mL) at room temperature under N_2 atmosphere. The ee value was determined by HPLC analysis. [b]MTBE as solvent. [c]t = 10 h. [d]t = 11 h. [e]t = 8 h.

carbonate (Table 2, entry 4), bulky 1-naphthoate (Table 2, entry 5), and 2-furanoate (Table 2, entry 6) were examined with no better enantioselectivity.

Next, the solvent effect was evaluated: PhMe, MeCN, dioxane, DME, CPME, and *c*-hexane gave no better results (Table 3, entries 1−3, 5−7). The reaction in MTBE afforded (*R*)-**3aa** in 94% ee with a yield of 55% (Table 3, entry 4). Furthermore, different palladium sources were also examined with THF as solvent. $Pd_2(dba)_3$ and $Pd(dba)_2$ made no big difference (Table 3, entries 9 and 10). $Pd(OAc)_2$, $PdCl_2$, and $[Pd(allyl)Cl]_2$ under the conditions of entry 8 exhibited a lower catalytic reactivity (Table 3, entries 11−13). Finally, we were pleased to find that, with a mixed solvent of MTBE and THF (4/1), the reaction afforded (*R*)-**3aa** in 70% NMR yield with 94% ee (Table 3, entry 14).

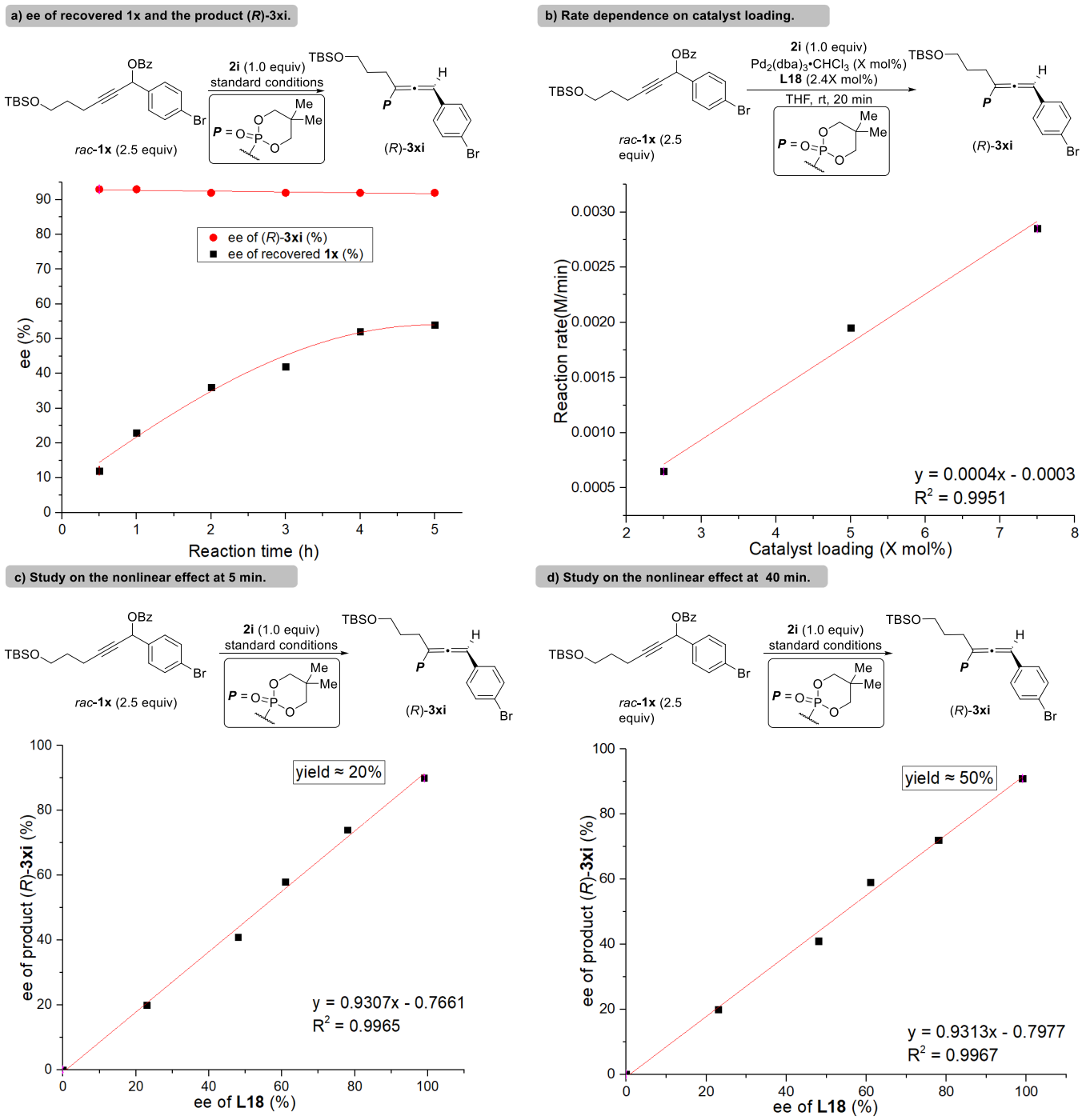
With the optimized conditions in hand (Table 3, entry 14), we next explored the scope of *H*-phosphonates (Scheme 2). When diethyl phosphite (**2a**) and dimethyl phosphite (**2b**) were used, the corresponding allenyl phosphonates (*R*)-**3aa** and (*R*)-**3ab** were obtained in decent yields and excellent ees. Dialkyl phosphites with longer alkyl chains also afforded the desired products (*R*)-**3ac**−(*R*)-**3ae** with excellent ees, but the yields were lower. No products were formed with diisopropyl phosphite (**1f**) and diphenyl phosphite (**1g**). The reaction of the 5-membered cyclic phosphite, 4,4,5,5-tetramethyl-1,3,2-dioxaphospholane 2-oxide (**2h**), with **1a** afforded (*R*)-**3ah** in 39% yield with 88% ee in MTBE. Delightfully, allenyl phosphonate (*R*)-**3ai** was obtained in 93% yield with 92% ee, and an *s* factor[18] of 44 was achieved when 6-membered 5,5-dimethyl-1,3,2-dioxaphosphorinan-2-one **2i** was applied at room temperature with THF as solvent.

We next turned our attention to the scope of propargylic benzoates (Scheme 3) with 5,5-dimethyl-1,3,2-dioxaphosphorinan-2-one (**2i**). Propargylic benzoates bearing a wide range of different Ar groups with either electron-donating (**1b**−**1d**) or electron-withdrawing substituents (**1e**-**1k**) all performed well under the standard conditions, giving the corresponding products (*R*)-**3bi**−(*R*)-**3ki** in good yields and with excellent enantioselectivities. 1-(Naphthyl)hept-2-ynyl benzoate (**1l**) also worked smoothly under the optimal conditions, forming

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Scheme 4. Mechanistic Studies



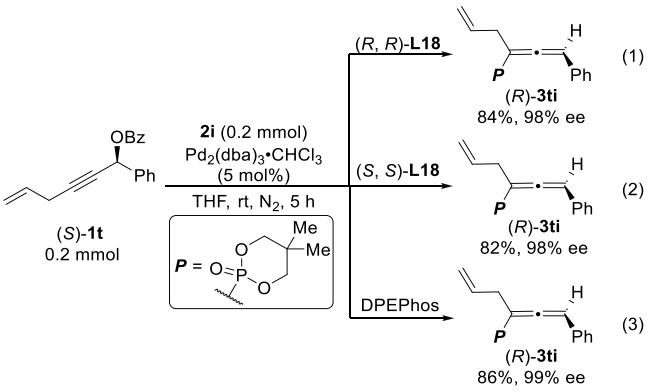
the allene (*R*)-**3li** in 80% yield with 91% ee. Then, an array of propargylic benzoates bearing different lengths of alkyl chains were examined, providing the corresponding products (*R*)-**3mi**−(*R*)-**3pi** in good yields and high enantioselectivities. Furthermore, propargylic benzoates bearing different alkyl group (R) groups with halogen (**1q**), allyl (**1t**), alkynyl (**1u**), cyano (**1v**), TBS-protected (TBS = *tert*-butyldimethylsilyl) alcohol (**1w** and **1x**), indole (**1y**), and carbazole (**1z**) were all compatible under the standard reaction conditions, which provide a handle for further transformations. The R group may also be an aryl group, giving the corresponding products (*R*)-**5ai**−(*R*)-**5di**. When the propargylic position was substituted with an alkyl group such as *n*-Pr, no desired allenyl phosphonate **5ei** was formed. Late-stage modification of bioactive complexes and pharmaceuticals such as cholesterol and adapalene derivatives afforded the corresponding allenes (*R*)-**3aai** and (*R*)-**3bbi** in good yields and excellent ee. The absolute configurations of the products were determined by single-crystal X-ray diffraction of (*R*)-**3hi** and (*R*)-**5ci**.

To unveil the reaction mechanism, the following experiments were conducted. First, the profile of the ee of the recovered (*R*)-**1x** and the product (*R*)-**3xi** vs time indicated that the reaction is of kinetic resolution (Scheme 4a). In addition, first-order dependence was observed for the catalyst obviously (Scheme 4b), and a linear relationship was observed between the ee of **L18** and product (*R*)-**3xi** at 5 or 40 min

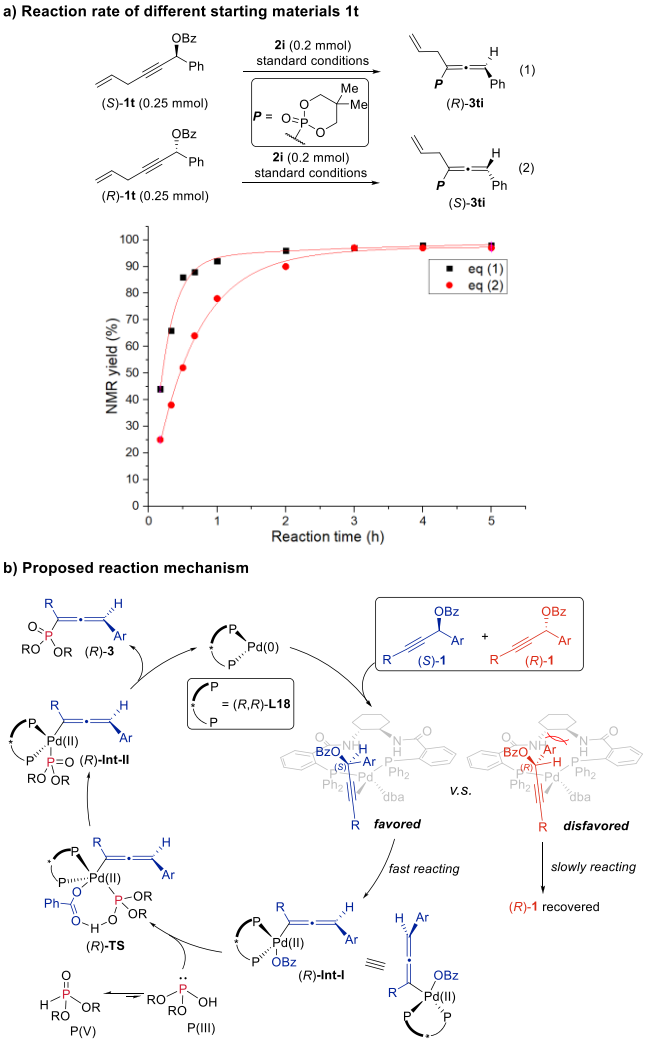
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Scheme 5. Investigating the Origin of Stereoselectivity

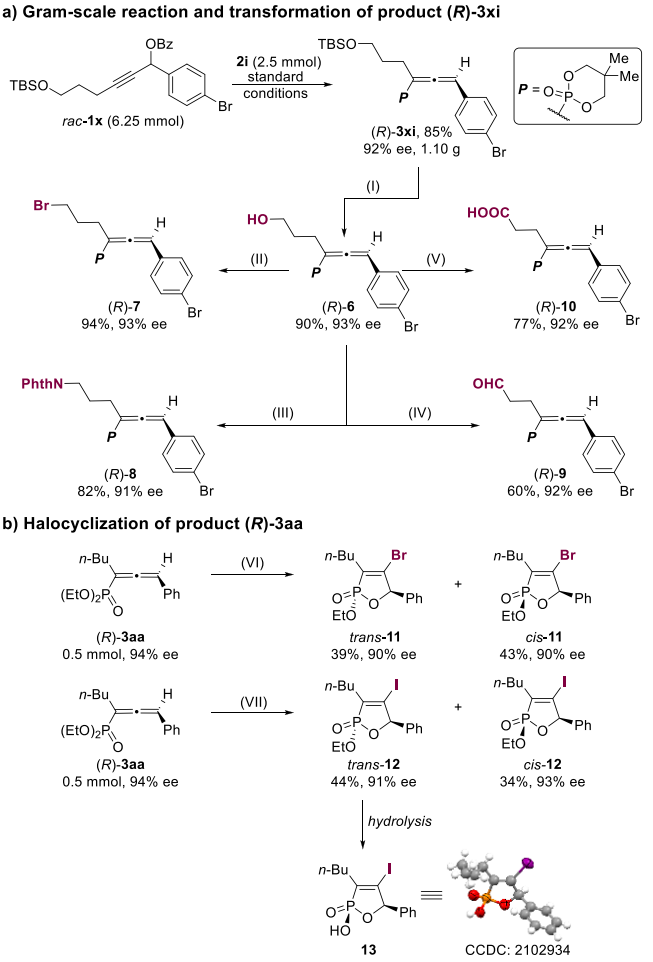


Scheme 6. Mechanistic Studies on Reaction Rates and Proposed Mechanism



(Scheme 4c,d), indicating that one ligand coordinated with one palladium atom, forming the active catalyst. In addition, the reaction of optically active (*S*)-**1t** under the standard conditions with (*R*,*R*)-**L18** as ligand afforded the product (*R*)-**3ti** in 84% yield with 98% ee (Scheme 5, eq 1). Surprisingly, we found that optically active (*S*)-**1t** under the standard conditions with (*S*,*S*)-**L18** or even DPEphos as ligand afforded the same product, (*R*)-**3ti**, in decent yields and excellent ees,

Scheme 7. Gram-Scale Reaction and Transformations of the Products[a]



[a]Reaction conditions: (I) HFPy (2.3 equiv), MeCN, 0 °C 10 min then rt 1 h. (II) CBr_4 (1.3 equiv), PPh_3 (1.5 equiv), DCM, 0 °C, 4 h. (III) PhthNH (1.0 equiv), PPh_3 (1.1 equiv), DIAD (1.2 equiv), THF, rt, 2 h. (IV) $Cu(NO_3)_2{\cdot}3H_2O$ (20 mol%), TEMPO (20 mol%), O_2 balloon, DCE, rt, 6.6 h. (V) $Fe(NO_3)_3{\cdot}9H_2O$ (25 mol%), TEMPO (25 mol%), KCl (25 mol%), O_2 balloon, DCE, rt, 13.8 h. (VI) Br_2 (1.5 equiv), MeCN/H_2O (1/4), 0 °C, 2 h. (VII) I_2 (1.1 equiv), MeCN/H_2O (1/5), rt, 7 h.

indicating the chiral catalyst was picking the matched enantiomer out of racemic propargylic benzoate (Scheme 5, eqs 2 and 3). Finally, it was observed that, with (*R*,*R*)-**L18** as the ligand, the reaction of (*S*)-**1t** was faster than that of (*R*)-**1t** under the standard reaction conditions (Scheme 6a).

Based on these results, we proposed the reaction mechanism as shown in Scheme 6b. First, chiral palladium(0) generated from the coordination of $Pd_2(dba)_3{\cdot}CHCl_3$ with **L18** underwent *anti*-oxidative addition[19] with propargylic benzoate **1** to deliver the allenyl palladium intermediate **I**. Then, the intermediate **II** was formed via the ligand exchange of intermediate **I** with the in situ-generated tautomeric P(III). Subsequent reductive elimination produced the final product and regenerated the chiral palladium(0). Due to the steric hindrance between the Ar substituent and the 1,2-diaminocyclohexane backbone of (*R*,*R*)-**L18**, the corresponding reaction of chiral palladium(0) species with (*S*)-**1** is faster than that with (*R*)-**1** (Scheme 6b), which resulted in high ee of the products (*R*)-**3**. The key here is that the coordination ability of

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the bidentated chiral ligand with Pd is by far much stronger than that of the in situ-generated trivalent P species.

To demonstrate the synthetic potential of this reaction, a gram-scale reaction was conducted to afford 1.10 g (85% yield) of the product (*R*)-**3xi** in 92% ee (Scheme 7a). The product (*R*)-**3xi** could be easily deprotected with HFPy at room temperature to afford allenol (*R*)-**6** in 90% yield with 93% ee;[20] Appel reaction of (*R*)-**6** gave allenyl bromide (*R*)-**7** in 94% yield with 93% ee;[21] *N*-allenyl amide (*R*)-**8** was obtained in 82% yield with 91% ee via Mitsunobu reaction of (*R*)-**6**;[22] allenyl aldehydes (*R*)-**9** was prepared from (*R*)-**6** in 60% yield and 92% ee using 20 mol% each of $Cu(NO_3)_2{\cdot}3H_2O$/TEMPO catalyst with an O_2 balloon;[23] allenyl carboxylic acid (*R*)-**10** was prepared from (*R*)-**6** in 77% yield and 92% ee using 25 mol% each of $Fe(NO_3)_3{\cdot}9H_2O$/TEMPO/KCl catalyst with an O_2 balloon.[24] Additionally, (*R*)-**3aa** was readily cyclized with Br_2 or I_2[25] to form the fully substituted valuable 1,2-oxaphosphol-3-ene 2-oxides *trans*-**11** (39%, 90% ee), *cis*-**11** (43%, 90% ee), *trans*-**12** (44%, 91% ee), and *cis*-**12** (34%, 93% ee); enantioenriched Br- or I-containing oxaphospholes *trans*-**11**/*cis*-**11** and *trans*-**12**/*cis*-**12** with potential biological activity[26] could be efficiently prepared from the allenyl phosphonate (*R*)-**3aa**. The absolute configurations of these products were determined by single-crystal X-ray diffraction of the hydrolyzed product **13** upon exposing *trans*-**12** to air for several days at ambient temperature according to the reported steric outcome of this step (Scheme 7b).[27]

In summary, we have presented the first example of catalytic asymmetric allenyl C−P bond formation reactions producing axially chiral allenylphosphonates from readily available propargylic benzoates. These products provide an efficient access to many synthetically and biologically useful phosphorus compounds. Mechanistic studies unveiled a kinetic resolution pathway. Further studies of related reactions are currently ongoing in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.2c04931.

Experimental procedures, characterization, and NMR spectra for obtained compounds (PDF)

Accession Codes

CCDC 2039512, 2082131 and 2102934 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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